WinLocalAssistant Proposed Changes and Validation Plan

v55.3.41 - defensive status and validation hardening

Purpose

This report consolidates the remaining user-requested WLA improvements, what has been implemented, what was corrected in the latest pass, and what should be tackled next. It is intended as a compact handoff/reference for future WLA work.

Implemented feature groups

- UI/audio: spoken audio, long-reply summarisation, mute controls, left rail, tutorial replay, changelog popup, dark/light setting and release checklist access.
- Quick commands: grouped quick controls moved out from the chat composer, with add/edit/move/delete/reset and browser persistence.
- Live Activity: taller resizable panel, indented subtasks, hover context, expandable details, and richer decision context.
- Project workflow: backend conversations, project memory, project-memory injection into build/edit prompts, file cards in replies and better project targeting checks.
- Validation: WLA Doctor, release gate, release checklist, Docker workspace validation, Docker patch validation, patch hash locking and patch promotion.
- Research: short synthesis added before raw source/claim mapping.
- Sandbox/image generation: Python-ready Linux sandbox default and honest image generation that refuses fake placeholders unless /image-draft is used.

Latest v55.3.41 fixes

- Fixed right-panel state mismatch where Working mode could still show Current task: Idle.
- Expanded /security-status into a defensive-only local inventory with Defender/AV/firewall/listening-port/Docker lab context.
- Added /security-triage and /defense-triage aliases.
- Added security status to the release checklist and integration radar.
- Enabled low-risk idle auto-apply by default while preserving strict gates around core files, network, install/update, rollback and high-risk changes.
- Updated Linux sandbox UI hints to prefer python --version after moving the default image to python:3.12-slim.

Important safety boundary

The security integration is defensive-only. It inventories local defensive tooling and suggests safe next steps. It does not automate offensive action, retaliation, unauthorised external scanning or exploitation. Kali tools remain for owned/local lab targets or authorised defensive analysis.

Validation ladder

- 1. Python compile: python -m compileall -q winlocalassistant tests_*.py app.py
- 2. Targeted pytest for the changed patch version.
- 3. Regression tests for the previous related patches.
- 4. Web UI JavaScript parse check when Node is available.
- 5. /wla-doctor and /release-gate final validation.
- 6. /docker-validate quick when Docker is available.
- 7. /docker-validate-patch before promoting any proposed patch.

Still recommended future work

- Add optional Playwright execution into /release-gate when WLA is running and Playwright is installed.
- Add a richer docked GUI release/security checklist panel instead of only popup/checklist output.
- Add project-target confirmation when a vague build request is made while a different project is currently open.
- Add documented read-only AVG/vendor CLI/API integrations where available before any user-approved remediation actions.
- Continue breaking down core.py into smaller modules with regression tests around every extraction.

Manual smoke commands

```
/wla-doctor
/release-checklist
/security-status
/security-triage
/integration-radar improve WLA reliability
/sandbox-linux stop
/sandbox-linux start
/sandbox-linux exec python --version
/image-generate realistic dachshund photo
/image-draft realistic dachshund photo
/fast make a tiny useful app
```